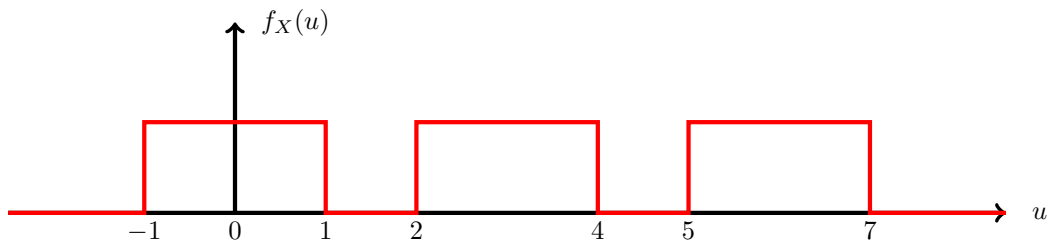


If random variable  $X$  has probability density function shown below, then what is the variance of  $E[X^2 - 2X + 1]$ ?

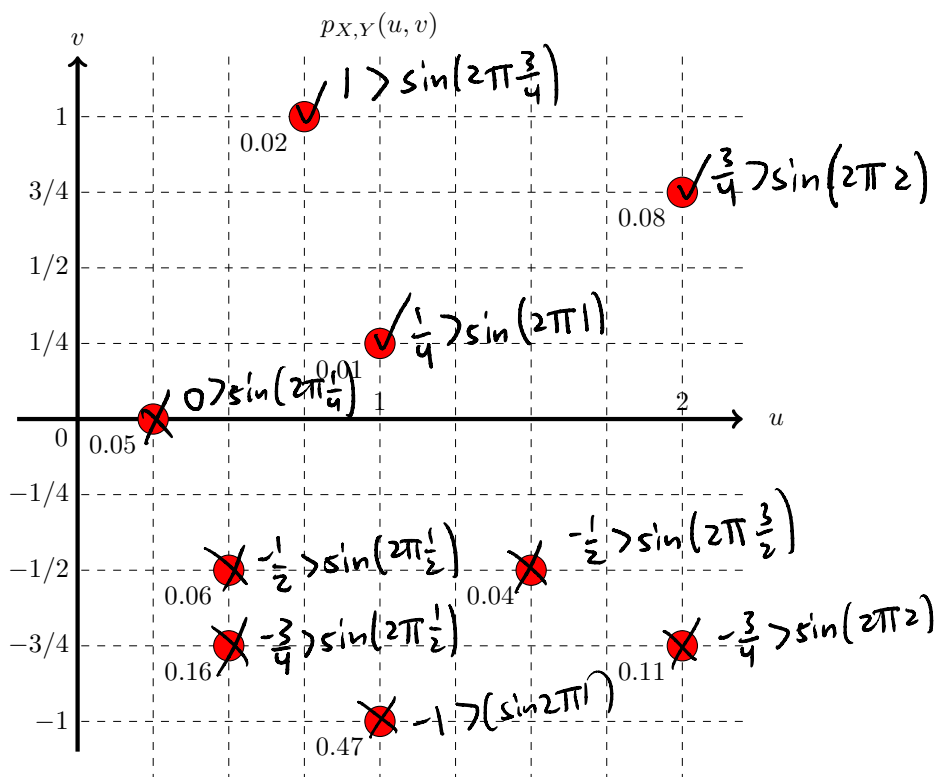


$$\text{Var}(E[X^2 - 2X + 1])$$

Variance of a constant is 0  
(EV is a constant)

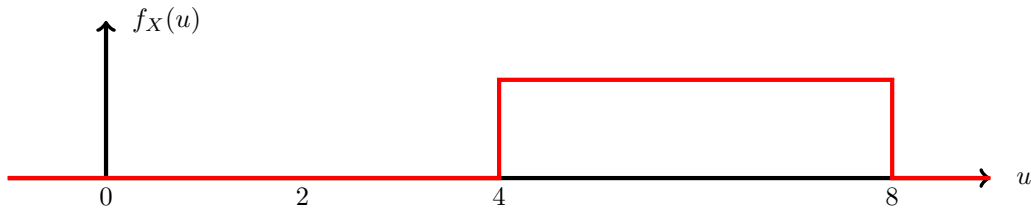
- (a) 0
- (b) 1
- (c) 2
- (d) 3
- (e) 4
- (f) 5
- (g)  $1/2$
- (h)  $3/2$
- (i)  $5/2$
- (j)  $31/3$
- (k)  $5734/45$
- (l) None of these

Random variables  $X$  and  $Y$  have the joint probability mass function shown below. What is the probability that  $Y$  is greater than  $\sin(2\pi X)$  ?



- (a) 0.11
- (b) 0.10
- (c) 0.02
- (d) 0.08
- (e) 0.01
- (f) 0.16
- (g) 0.15
- (h) 0.17
- (i) 0.22
- (j) 0.26
- (k) 0
- (l) None of these
- $P(Y > \sin(2\pi X)) = 0.02 + 0.08 + 0.01 = 0.11$

If random variable  $X$  has probability density function shown below, then what is the expected value of  $\sqrt{X}$ ?



$$1 = 4h \rightarrow h = \frac{1}{4}$$

$$f_X(u) = \begin{cases} \frac{1}{4}, & 4 \leq u \leq 8 \\ 0, & \text{else} \end{cases}$$

(a)  $4(2\sqrt{2} - 1)/3$

(b)  $16(2\sqrt{2} - 1)/3$

(c)  $(2\sqrt{2} - 1)/3$

(d)  $4(\sqrt{2} - 1)/3$

(e)  $4(2\sqrt{2} - 1)$

(f)  $4(2\sqrt{2} + 1)/3$

(g)  $8(2\sqrt{2} - 1)/3$

(h) 2

(i)  $5/2$

(j)  $2\sqrt{2}$

(k)  $4\sqrt{2}$

(l) None of these

$$E[g(x)] = \int_{-\infty}^{\infty} g(u) f(u) du$$

$$\rightarrow E[\sqrt{X}] = \int_{-\infty}^{\infty} \sqrt{u} f_X(u) du$$

$$= \int_4^8 \sqrt{u} \frac{1}{4} du$$

$$= \frac{1}{4} \left[ \frac{2}{3} u^{\frac{3}{2}} \right]_4^8$$

$$= \frac{1}{6} \left[ 8^{\frac{3}{2}} - 4^{\frac{3}{2}} \right]$$

$$= \frac{1}{6} \left[ \sqrt{8}^3 - \sqrt{4}^3 \right]$$

$$= \frac{1}{6} \left[ 2\sqrt{2}^3 - 8 \right]$$

$$= \frac{1}{6} \left[ 16\sqrt{2} - 8 \right] = \frac{4}{3} (2\sqrt{2} - 1)$$

$$u^{\frac{1}{2}} \rightarrow \frac{2}{3} u^{\frac{3}{2}}$$